

Overall Comparison of Sentiment Analysis Methods

The objective of this project was to compare multiple approaches for sentiment classification on the TweetEval sentiment dataset from the Cardiff NLP research group, ranging from lexicon-based methods and classical machine learning algorithms to deep learning architectures and transformer-based language models. Table 1 summarizes the final results obtained by all evaluated methods.

Table 1: Comparison of all evaluated models.

Category	Model	Method	Accuracy (%)	Train (s)
Lexicon-Based	VADER	NLTK	55.00	N/A
Lexicon-Based	TextBlob	TextBlob	50.00	N/A
Rule-Based	Custom Rules	Custom	51.00	N/A
Classical ML	Logistic Regression	TF-IDF	62.20	0.8
Classical ML	SVM	TF-IDF	62.24	0.4
Classical ML	XGBoost	TF-IDF	59.51	23
Deep Learning	LSTM	Keras	45.31	52
Deep Learning	ED-LSTM	Keras	45.58	54
Deep Learning	Bi-LSTM	Keras	63.70	102
Transformer	DistilBERT-ML	Full FT	59.73	449
Transformer	DistilBERT-ML	LoRA	62.22	131
Transformer	mBERT	Full FT	62.69	706
Transformer	mBERT	LoRA	63.45	262
Transformer	XLM-RoBERTa	Full FT	65.39	913
Transformer	XLM-RoBERTa	LoRA	65.96	261

The experimental results revealed substantial differences between the evaluated sentiment analysis approaches. The comparison included lexicon-based methods, rule-based systems, classical machine learning algorithms, recurrent neural networks, and transformer-based language models. Overall, a clear progression in predictive performance can be observed as model complexity increases.

The weakest baselines were provided by the lexicon-based and rule-based approaches. VADER achieved the highest accuracy among these methods, reaching 55.00%, while the custom rule-based system and TextBlob obtained accuracies of 51.00% and 50.00%, respectively. Although these approaches require no training and are computationally inexpensive, their limited ability to capture contextual information significantly restricts their predictive performance.

A substantial improvement was achieved by introducing supervised machine learning models based on TF-IDF features. Both SVM and Logistic Regression reached accuracies above 62%, outperforming all lexicon-based approaches by a considerable margin. Remarkably, these models required less than one second of training time, making them extremely efficient solutions. XGBoost achieved a slightly lower accuracy of 59.51% while requiring approximately 23 seconds of training.

Among the deep learning architectures, Bi-LSTM achieved the strongest result with an accuracy of 63.70%, performing competitively against several transformer-based models. This demonstrates the benefits of bidirectional sequence modeling for sentiment classification. In contrast, the standard LSTM and Encoder-Decoder LSTM architectures achieved accuracies of only 45.31% and 45.58%, respectively, making them the weakest machine learning models evaluated in this project.

The highest overall performance was achieved by transformer-based models. The best result was obtained by XLM-RoBERTa with LoRA fine-tuning, which reached an accuracy of 65.96%. Interestingly, this model slightly outperformed its fully fine-tuned counterpart, which achieved 65.39% accuracy despite requiring substantially more training time (913 seconds compared to 261 seconds). Similar behavior was observed for the other transformer architectures, where LoRA consistently achieved performance comparable to full fine-tuning while requiring significantly fewer computational resources.

It should be noted that the transformer models were trained on a subsampled version of the training dataset in order to reduce computational requirements and training time. Despite this limitation, XLM-RoBERTa remained the strongest model among all evaluated approaches. This suggests that additional performance improvements could potentially be achieved through longer training schedules or by utilizing the complete training dataset.

One of the most interesting findings of this project is the relationship between predictive performance and computational cost. While transformer models achieved the highest accuracies, the improvement over the best classical machine learning methods was relatively modest. For example, XLM-RoBERTa with LoRA outperformed SVM by approximately 3.7 percentage points, but required hundreds of times more training time. This highlights the importance of considering computational efficiency alongside predictive performance when selecting a model for practical applications.

Overall, the results demonstrate a clear progression from lexicon-based methods, through classical machine learning and deep learning approaches, to transformer architectures. XLM-RoBERTa with LoRA fine-tuning emerged as the most attractive solution, providing the highest accuracy while maintaining a reasonable computational cost. At the same time, the strong performance and extremely short training times of SVM and Logistic Regression indicate that classical machine learning methods remain highly competitive baselines for sentiment classification tasks.